

Junxian Li

CONTACT INFORMATION	University of California, Davis Department of Mathematics One Shields Avenue Davis, CA 95616	njxli@ucdavis.edu Homepage Google Scholar
RESEARCH INTERESTS	Analytic Number theory: L -functions, Primes, Diophantine Equations, Exponential Sums Automorphic Forms, Dynamical systems, Additive Combinatorics	
EMPLOYMENT	Assistant Professor University of California, Davis	2023–Now
	Postdoctoral Researcher (Mentor: Valentin Blomer) Universität Bonn Max Planck Institute for Mathematics Georg-August Universität Göttingen	2021–2023 2019–2021 2018–2019
EDUCATION	Ph.D. in Mathematics (Advisor: Alexandru Zaharescu) University of Illinois at Urbana-Champaign	2013–2018
	B. A. in Mathematics Nanjing University	2009–2013
PUBLICATIONS	24. A higher rank shifted convolution problem with applications to L-functions (with V. Blomer), arXiv:2511.03294, 29 pp. 23. Level aspect subconvexity for $GL(2) \times GL(2)$ (with K. Aggarwal, S. Kumar, C.-H. Kwan, W. H. Leung, and Matthew P. Young), arXiv:2412.12410, 27 pp. 22. A two dimensional version of the delta symbol method (with S. Rydin Myerson and P. Vishe), arXiv:2411.11355, 74 pp. 21. Joint value distribution of L-functions on the critical line (with S. Inoue), arXiv:2102.12724, 50 pp. 20. Simultaneous large values and dependence of Dirichlet L-functions in the critical strip (with S. Inoue), J. Lond. Math. Soc. 112 (4), e70322, arXiv:2211.15165, 24 pp, . 19. Simultaneous extreme values of zeta and L-functions (with W. Heap), Math. Ann. 390, 6355–6397, 2024. 18. Correlations of values of random diagonal forms (with V. Blomer), Int. Math. Res. Not., 2023 (23), 20296–20336. 17. Additive problems with almost prime squares (with V. Blomer, L. Grimmelt and S. Rydin Myerson), Geom. Funct. Anal., 33 (2023), 1173–1242. 16. Lower bounds for discrete negative moments of the Riemann zeta function (with W. Heap and J. Zhao), Algebra Number Theory 16 (2022), no. 7, 1589–1625.	

15. Large values of Dirichlet L -functions at zeros of a class of L -functions, **Canad. J. Math.** *73* (2021), no. 6, 1459–1505.
 14. Uniform Titchmarsh divisor problems (with E. Assing and V. Blomer), **Adv. Math.** *393* (2021), Paper No. 108076, 51 pp.
 13. Leading Digits of Mersenne Numbers (with Z. Cai, M Faust, A. J. Hildebrand, and Y. Zhang), **Exp. Math.** *30* (2021), no. 3, 405–421.
 12. The surprising accuracy of Benford’s law in mathematics (with Z. Cai, M. Faust, A. J. Hildebrand and Y. Zhang), **Amer. Math. Monthly** *127* (2020), no. 3, 217–237. (*winner of the Paul R. Halmos-Lester R. Ford Award*)
 11. Ducci iterates and similar ordering on sets of visible points (with A. Tamazyan and A. Zaharescu), **Int. J. Number Theory** *16* (2020), no. 1, 1–28.
 10. A binary quadratic Titchmarsh divisor problem **Acta Arithmetica** *192* (2020), no. 4, 341–361.
 9. The final problem: an identity from Ramanujan’s lost notebook (with B. Berndt and A. Zaharescu), **J. Lond. Math. Soc.** (2) *100* (2019), no. 2, 568–591.
 8. Almost Beatty Partitions (with A. J. Hildebrand, X. Li, and Y. Xie), **J. Integer Seq.** *22* (2019), no. 4, Art. 19.4.6, 34 pp.
 7. Value distribution of $L'(\rho)$ (with A. Zaharescu), **J. Math. Anal. Appl.** *480* (2019), no. 1, 123400, 24 pp.
 6. A local Benford Law for a class of arithmetic sequences (with Z. Cai and A. J. Hildebrand), **Int. J. Number Theory** *15* (2019), no. 3, 613–638.
 5. On distinct consecutive r -differences (with G. Shakan), **J. Number Theory** *199* (2019), 363–376.
 4. Exact evaluation of second moments associated with some families of curves over a finite field (with R. Donepudi and A. Zaharescu), **Finite Fields Appl.** *48* (2017), 331–355.
 3. A lower bound for the least prime in an arithmetic progression (with K. Pratt and G. Shakan), **Q. J. Math.** *68* (2017), no. 3, 729–758.
 2. Smooth L^2 distances and zeros of approximations of Dedekind zeta functions (with M. Nastaescu, A. Roy, and A. Zaharescu), **Manuscripta Math.** *154* (2017), no. 1-2, 195–223.
 1. Zeros of a family of approximations of Hecke L -functions associated with cusp forms (with A. Roy and A. Zaharescu), **Ramanujan J.** *41* (2016), no. 1-3, 391–419.
-
- CONFERENCE
PROCEEDINGS

3. Shifted convolution sums and L -functions, **Oberwolfach Report** 51 (2025), 2751–2753.
 2. The Final Problem: A Series Identity from the Lost Notebook (with B. C. Bruce and A. Zaharescu), **George E. Andrews 80 Years of Combinatory Analysis**, K. Alladi, B. C. Berndt, P. Paule, J. Sellers, and A. J. Yee, eds., Birkhäuser, 783–790, 2021.
 1. On primes in arithmetic progressions Automorphic forms and related topics, **Contemp. Math.** 732, 165–167, Amer. Math. Soc., Providence, RI, 2019.

GRANTS AND AWARDS	• NSF Grant (DMS-2502537)	2025–2027
	Arithmetic, L-functions and Automorphic Forms	
	• UC Davis Hellman Fellowship	2025–2026
	for potential for great distinction in research	
	• The Paul R. Halmos-Lester R. Ford Award	2021
	for outstanding expository papers in The American Mathematical Monthly	
	• Bateman Fellowship	2018
	for excellence in Number Theory	
	CONFERENCE AND TRAVEL GRANTS	
	• Conference Grants	
	– Bonn International Graduate school of Mathematics	Mar 2022
	• Travel Grants	
	– AMS-Mathematics Research Communities Grant for the JMM	Jan 2019
	– US Junior Oberwolfach Fellows	Oct 2017
	– PCMI-Graduate Summer School Travel Award	June 2017
	– UIUC-Association for Women in Mathematics Graduate Travel Funding	2017–2018
	– AMS Graduate Student Travel Grant for the JMM	Jan 2017
TEACHING	• Graduate Courses and Seminars	
	– MAT 280 Number Theory, Instructor	Davis, 2025W
	– MAT 298 Group Study, Instructor	Davis, 2024-Now
	– Topic course: Sieve Methods, Instructor	Bonn, 2022
	– Number Theory Learning Seminar	Göttingen, 2018-2019
	• Undergraduate Courses	
	– MAT 127B Real Analysis, Instructor	Davis, 2027W
	– MAT 127A Real Analysis, Instructor	Davis, 2026F
	– MAT 199 Undergraduate Research	Davis, 2026S
	– MAT 115 Number Theory, Instructor	Davis, 2024F, 2025F
	– MAT 148 Discrete Math, Instructor	Davis, 2024W
	– MAT 21A Calculus I, Instructor	Davis, 2023F, 2024F, 2026W, 2027W
	– Math 415 Linear Algebra, Instructor	UIUC, 2017
	• On the List of Teachers Ranked as Excellent by their Students	Fall 2017
	– Math 231 Calculus II, Instructor	UIUC, 2016
	– Math 241 Calculus III, Instructor	UIUC, 2016–2015
	• Graduate supervision	
	– Adrian Weitzer (PhD advisee, UC Davis)	2025–Now
	– Thomas Allen (Qualifying and Dissertation Committee, UC Davis)	2026
	– Colby Brown (Dissertation Committee, UC Davis)	2026
	– Ivan Chan Kai Chin (Universität Bonn)	2022–2023
STUDENT MENTORING	• Undergraduate supervision	
	– Ruihan (Robin) Ma, David Reagon, Yu (Wilson) Wen (UC Davis)	2026–Now
	– Jiacheng Sun (UC Davis)	2025–2026
	Undergraduate Students in Illinois Geometry Lab (10 projects)	
	– Almost Beatty Partitions; Beatty sequences, and Partitions of the Integers	2018
	– Chaotic maps and exotic number systems	Fall 2017
	– Finding integers in group orbits	Spring 2017
	– Local Benford’s Law; Leading digit distribution	2016
	– Fractals, Patterns and Randomness in Number Theory	2015
	– Fourier Series with Number theoretic coefficients	Fall 2014
	– Symmetry in Nature	Spring 2014
PROFESSIONAL SERVICES	CONFERENCE ORGANIZATION	

- **Co-organizer of BIRS workshop** Sept 2026
– Arithmetic, L-functions, and pseudorandomness
- **Co-organizer of a workshop at Universität Bonn** 2022
– Young Scholars in the Analytic Theory of Numbers and Automorphic Forms
- **Co-organizer of AMS Special Session at the Joint Mathematics Meeting** 2019
– Number Theoretic Methods in Hyperbolic Geometry

SEMINAR ORGANIZATION

- **Co-organizer of AG-NT seminar at UC Davis** 2023–2025
- **Organizer of Graduate Student Number Theory Seminar in UIUC** 2016–2018

STUDENT MENTORING

- **Founder of Davis Math Lab Program** Mar 2026–Now
– undergraduate research supervised by faculty and graduate mentors
- **Lecturer for AWM-Math Club Lecture Series (2)** Feb 2024
- **Panelist of Graduate Research Opportunities for Women at Bonn** Mar 2023

REFEREE

- J. Inst. Math. Jussieu; Crelle; Algebra Number Theory; Math. Z.; Forum Math.;
- Canad. J. Math.; Res. Number Theory; Monatsh. Math.; Bull. London Math. Soc.
- Bull. Sci. Math.; Acta Math. Hung.
- Ramanujan J.; J. Number Theory; Indag. Math. (N.S.); Acta Arith.
- Math. Reports; Rev. Roumaine Math. Pures Appl.; J. Math. Sci. Adv. Appl.

DEPARTMENTAL SERVICE

- Qualifying Exam and Thesis Committees 2026
- KAP Search Committee 2025
- Faculty Representative Committee 2024–2026
- Departmental Relations Committee 2023

INVITED TALKS IN CONFERENCES AND SEMINARS

- **Moments of Twisted $GL(3)$ L -functions**
 - Pan Asia Number Theory (PANT) conference, NU Singapore Nov, 2026
 - FOUVRY 73, the EPFL Bernoulli Center Aug, 2026
 - Probability in Number Theory Workshop, CRM Montreal June, 2026
 - Xi'an Number Theory Forum, (online) June, 2026
 - Recent Perspectives on Moments of L -Functions, UT Dallas June, 2026
- **Variation of the Delta Symbol Method**
 - PLeaSANT, MPIM, Bonn June, 2026
 - Number Theory Seminar, Rutgers Apr, 2026
- **Shifted convolution sums and L -functions**
 - Number Theory Web Seminar, (online) Jan, 2026
 - AGNT Seminar, UC Davis Jan, 2026
 - Analytic Number Theory, Oberwolfach Nov, 2025
- **Subconvexity of Rankin-Selberg L -functions**
 - UVA Number Theory Seminar Oct, 2025
 - Chinese Academy of Sciences Sept, 2025
 - UNSW Number Theory Days, Sydney (online) Aug 2025
 - Progress and Emergent Theories in Zeta and L-Functions, MPIM, Bonn July 2025
 - Full circle: 100 years of the circle method, Institut Mittag-Leffler July 2025
 - JMM, Seattle Jan 2025
- **Nonvanishing of twists of L -functions**
 - Analytic Number Theory Program, Mittag Leffler Apr 2024
- **Correlations of random quadratic forms**
 - Antha-PDE, University of Warwick July 2023
- **Two dimensional Kloosterman's refinement of the circle method**
 - Chinese Academy of Sciences Sept, 2025

– Prime numbers and arithmetic randomness, CIRM	June 2025
– Number Theory Seminar, Purdue	Apr 2024
– Number Theory Seminar, Texas A&M	Apr 2024
– Number Theory Seminar, UC Irvine	Feb 2024
– Number Theory Seminar, Stanford	Nov 2023
– Delta Method and Subconvexity, AIM	Oct 2023
– MAGNTS, Ann Arbor	Oct 2023
– Number Theory Seminar, Lille	Sept 2022
– Oberseminar Analytic Number Theory and Automorphic Forms, Bonn	Apr 2022
• Hardy-Littlewood problems with almost primes	
– Analytic Number Theory Meetings, IHP.	Sept 2022
– Number Theory Days, HKU (online).	July 2022
– Workshop in Number theory and Harmonic Analysis, SDU (online).	July 2022
– Number Theory Seminar, UIUC (online).	Mar 2022
– Heilbronn Number Theory Seminar, Bristol (online).	Jan 2022
– Number Theory Seminar, XJTU (online).	Dec 2021
• Simultaneous large values of Dirichlet L-functions in the critical strip	
– Oberseminar Analytic Number Theory and Automorphic Forms, Bonn.	Oct 2021
• Joint Value distribution of L-functions	
– Number Theory Seminar, PIMS Collaborative Research Group (online).	Sept 2022
– Qilu Youth Forum, SDU (online).	Sept 2021
– Number theory lunch seminar, MPIM (online).	Sept 2021
• Uniform Titchmarsh Divisor Problems	
– ArStAFANT Workshop, EPF Lausanne.	June 2023
– Séminaire ADA, Calais.	Sept 2022
– Number theory Seminar, SDU (online).	May 2021
– PIMS-Lethbridge Number Theory Seminar, Lethbridge (online).	Mar 2021
– Japan Europe Number Theory Exchange Seminar.	Jan 2021
• Joint Value Distribution of L-functions.	
– Oberseminar Analytic Number Theory, Bonn (online).	Nov 2020
• Derivative of the Riemann zeta function at its zeros.	
– Analytic Number Theory Meeting, IHP (online).	June 2020
• Extreme values of L-functions	
– Number theory lunch seminar, MPIM.	Oct 2019
– Oberseminar number theory, Georg-August Universität Göttingen.	Nov 2018
• The Unreasonable Effectiveness of Benford’s Law in Mathematics	
– Joint with A. J. Hildebrand, Number Theory Seminar, UIUC.	Apr 2018
• Primes in arithmetic progressions	
– Junior Mathematics Colloquium, Georg-August Universität Göttingen.	Dec 2017
• Randomness in Number Theory	
– Graduate Student Colloquium, UIUC.	Nov 2017
• Primes in arithmetic progressions	
– Where Geometry meets Number Theory, a conference in honor of the 60th birthday of Per Salberger, Gothenburg.	July 2017
• The least prime in an arithmetic progression	
– Joint Mathematics Meeting, Atlanta.	Jan 2017
– Number Theory Seminar, UIUC.	Sept 2016
– Workshop on Automorphic Forms and Related Topics, Sarajevo .	Jul 2016
• Approximations of L-functions	
– Midwest Number Theory Conference for Graduate Students and Recent Ph.D’s.	Oct 2015
– Graduate Student Number Theory Seminar, UIUC.	Nov 2015
• Bailey Pairs and Bailey chains	
– q -series Seminar, UIUC.	Apr 2015
• Basic Hypergeometric functions	

CONFERENCES
AND SUMMER
SCHOOLS

- Analytic Number Theory Conference, CIRM Nov, 2027
- Pan Asia Number Theory (PANT) conference, NU Singapore Nov, 2026
- FOUVRY 73, the EPFL Bernoulli Center Aug, 2026
- Probability in Number Theory Workshop, CRM June, 2026
- Recent Perspectives on Moments of L -functions June, 2026
- Universal Statistics in Number Theory Summer School, CRM May, 2026
- Arithmetic Statistics for Algebraic Objects, Oberwolfach Nov 2025
- Analytic Number Theory, Oberwolfach Nov 2025
- Aarhus Automorphic Forms, Aarhus Aug 2025
- Progress and Emergent Theories in Zeta and L-Functions, MPIM, Bonn July 2025
- Full circle: 100 years of the circle method, Institut Mittag-Leffler July 2025
- Prime numbers and arithmetic randomness, CIRM June 2025
- Harmonic analysis and analytic number theory, Hausdorff Institute, Bonn April 2025
- Visions in Arithmetic & Beyond: Celebrating Peter Sarnak's Work and Impact June 2024
- International conference on L -functions and automorphic forms May 2024
- BIRS-CMI workshop: New Directions in Rational Points, Chennai Jan 2024
- Delta symbols and the subconvexity problem, AIM Oct 2023
- RHB70: Analytic Number Theory and Its Interfaces, Oxford July 2023
- Women and Mathematics, IAS May 2023
- Analytic Number Theory Workshop, Oberwolfach Nov 2022
- 50 years of Number Theory and Random Matrix Theory, IAS June 2022
- Harmonic Analysis and Number Theory, ETH Mar 2022
- Zeta functions, CIRM Dec 2019
- Second Symposium on Analytic Number Theory, Cetraro July 2019
- Rational points on irrational varieties, IHP June 2019
- L-functions and Multiplicative Number Theory, U of Mississippi May 2019
- Distribution of values of zeta functions and L-functions, RIKEN Mar 2019
- Workshop and Winter School on Local Statistics of Point Sequences, Linz Feb 2019
- Building Bridges: 4th EU/US Summer School
and Workshop on Automorphic Forms and Related Topics July 2018
- Hausdorff School: L-functions: Open Problems and Current Methods June 2018
- MRC: Number Theoretic Methods in Hyperbolic Geometry June 2018
- Probability in Number Theory May 2018
- Arbeitsgemeinschaft in Oberwolfach Oct 2017
- MSRI Summer Graduate School on Automorphic Forms
and the Langlands Program Aug 2017
- PCMI Graduate Summer School on random matrices June 2017
- University of Houston Summer School on Dynamical Systems May 2017
- MSRI: Analytic Number Theory Jan, May 2017
- West Coast Algebraic Topology Summer School Aug 2016
- Building Bridges: 3rd EU/US Summer School
and workshop on Automorphic Forms July 2016
- UNCG Summer School in Computational Number Theory June 2016
- Houston Summer School on Dynamical Systems May 2016
- UNCG Summer School in Computational Number Theory May 2015
- Exchange in University of Wisconsin-Madison Fall 2012

OUTREACH
ACTIVITIES

- Organizer of Lunar New Year Celebration Davis, Feb 12, 2024
- Four Color Fest Illinois, Nov 1-4, 2017
- A Math Carnival at Illinois-Gathering for Gardener Illinois, Jan 28, 2017
- Science at the Market Illinois, Aug 2013